

Emphasis of design

As in any other product, design plays a pivotal role in the success of robotic kits. The design is dependent on the target group. Kids want their kits to be colourful, easy to use and robust. Engineering students want the circuits to be open, the kit to be modular and facilities of add-on modules, whereas lab equip-



Cyton I robot manipulator from Energid Technologies

Hardware structure of certain models can be assembled in multiple ways, so that you can construct different robots using the same parts. —Thushar V. Nadh, Director-Technical, Rhydo Technologies



ment should be robust and critical and circuits should be closed and secured.

The other design considerations are design efficiency, economy, material/component availability, optimising manufacturing process, etc. If the design is targeted for Indian customers, the emphasis is on value for money. For international clients, the emphasis is on kits that solve complex problems.

Cost of the robot can be reduced by changing the materials used in construction of body parts. Manufacturers study the end product and select the actuators and sensors that are best

suited for the robot. For example, certain robots need precision rather than speed. In that case, a stepper motor is used instead of servo or brushless DC motors.

Cost has a direct relationship with quantity. In India, the demand in this segment is very low compared to other countries and hence the costs are quite high. However, with an increase in awareness and hence demand, prices of robotic kits will fall.

To make the kits appealing to the Indian consumers, manufacturers need to come up with different kinds of



sleuthHound automatic locking and tracking camera system from Gridbots



BOOKS

NEW RELEASE

Power Electronics

Devices Circuits and MATLAB Simulations

Power Electronics
Devices Circuits and
MATLAB Simulations

Includes CD

Alok Jain
Price: Rs. 315/-

Fundamentals of Microcontrollers and Applications in Embedded Systems



Includes CD

Ramesh Ganurkar
Price: Rs. 300/-

30% Discount

Parallel Port Complete



Includes CD

Jan Axelsson
Price: Rs. 330/-

45% Discount
on following two books till stock lasts

USB Complete



Includes CD

Third Edition
Jan Axelsson
Price: Rs. 380/-

Serial Port Complete



Includes CD

Jan Axelsson
Price: Rs. 330/-

Programmable Logic Controllers & Industrial Automation: An Introduction



Madhuchandra Mitra
Samarjit Sen Gupta
Price: Rs. 200/-

Embedded Ethernet & Internet Complete



Includes CD

Jan Axelsson
Price: Rs. 390/-

Illustrated Programming with C++



Includes CD

Dilip Mali
Pranod Vasankar
Price: Rs. 210/-

C Application Programs and Projects



Includes CD

Dilip Mali
Pranod Vasankar
Price: Rs. 150/-

Elements of Artificial Neural Networks



2nd Edition
K. Mehrotra, C. Mohan, Sanjay Ranka
Price: Rs. 320/-

The Microcontroller Idea Book



Jan Axelsson
Price: Rs. 240/-

Penram International Publishing (India) Pvt. Ltd.
B-28, Basse Udyog Bhawan, I.I. Road, Sector 14, Gurgaon - 122 002
Tel: 91-22-68624247 / 68624248 : Fax: 91-22-2448065
Email - penram@penram.com • Website - www.penram.com